

Mixed Holomorphic–Topological Strings

The Companion

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Abstract

The local Bershadsky–Cecotti–Ooguri–Vafa field theory on $\mathbb{R}_{\text{top}}^2 \times \mathbb{C}_{\text{hol}}^2$, truncated to its Hamiltonian BF sector, poses at each brane point a mixed holomorphic–topological Deligne problem: compare the bulk factorization algebra with the Morita E_2 -centre of the boundary category. This volume proves exactly one formal-local stalk of that problem, and maps with type-theoretic precision everything that separates the stalk from the global theory. The proved stalk: at N Dirac branes the closed local algebra is the continuous Chevalley–Eilenberg algebra of $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}_{\text{cont}}^\vee[1]$, $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}$, the boundary chart is the Koszul algebra of the derived commuting variety, and the dictionary $c_f \mapsto \theta_f$, $u_f \mapsto J(f) = \text{Tr } f(\phi_1, \phi_2)$ is the unique formal-Darboux coordinate of the scalar-reduced stable trace sector. The removed scalar line returns as the Capelli anomaly $\hbar_{\mathbb{W}} N[\bar{c}]$. This companion states the stalk theorem, the uniqueness axioms, the completion analysis with its proved obstructions, the typed ledger, and the three conjectures that point at the compact world -- ending with the operator lift of the Igusa form.

I The seed

A holomorphic–topological theory in complex dimension two is the simplest habitat where the chiral world of the curve volumes meets the Calabi–Yau world of the threefold volumes: two real topological directions, two holomorphic ones, and at a worldline defect the question every holography asks -- do the bulk observables equal the centre of the boundary? The volume’s discipline is to refuse the question in the large and answer it exactly in the small. In the small: at a formal holomorphic-symplectic brane point of $X = \mathbb{R}_{\text{top}}^2 \times \mathbb{C}_{\text{hol}}^2$ with symplectic form $dz_1 \wedge dz_2$, the Hamiltonian vector fields form $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ with bracket $\{f, g\} = \partial_{z_1} f \partial_{z_2} g - \partial_{z_2} f \partial_{z_1} g$; the shifted-cotangent BF theory has fields valued in $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}_{\text{cont}}^\vee[1]$; and N Chan–Paton branes present the boundary as the derived commuting variety $[\mu_{\text{der}}^{-1}(\mathfrak{o})/GL_N]$, $Q\psi = [\phi_1, \phi_2]$. The trace map

$$J(f) = \overline{\text{Tr } f(\phi_1, \phi_2)}$$

is the whole local dictionary: it is the formal-Darboux coordinate of the scalar-reduced stable trace sector, it satisfies $\{J(f), J(g)\} = J(\{f, g\}) + N[\{f, g\}]_{\mathfrak{o}}$, and the scalar defect $[\{f, g\}]_{\mathfrak{o}}$ -- the constant the quotient by $\mathbb{C} \cdot 1$ discarded -- returns as a projective anomaly, the Capelli scalar $\hbar_{\mathbb{W}} N[\bar{c}]$: on linear generators, $\{\text{Tr } \phi_1, \text{Tr } \phi_2\} = N$. Nothing else about the global theory is claimed. The volume’s real product is the honest map: a typed ledger of the obstructions -- descent, quantum master equation,

anomaly, locality, centre lift -- that separate this stalk from a compact target, so that every global statement in the constellation must either supply the named data or remain a criterion.

2 The objects

Definition 2.1 (The local pair). *Closed: the continuous Chevalley–Eilenberg algebra $C_{\text{CE},\text{cont}}^\bullet(\mathfrak{g})$, with $\mathfrak{h}_{\text{cont}}^\vee$ the Grothendieck–Matlis residue dual spanned by $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$ with $\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{am} \delta_{bn}$. Open: $A_{\partial,N}^{\text{cl}} = C_{\text{CE}}^\bullet(\mathfrak{gl}_N, \text{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1]))$ with Koszul differential $Q\psi = [\phi_1, \phi_2]$; the moment map is traceless, so the effective presentation replaces $\mathfrak{gl}_N[1]$ by $\mathfrak{sl}_N[1]$. Stable: the trace algebra $A_{\partial,H}^{\text{st}} \cong \widehat{\mathbf{S}}_{\text{Tate}}(\mathfrak{h})$, reached by scalar reduction, degreewise stable Procesi–Razmyslov trace passage, and the Roos limit convention.*

Definition 2.2 (The admissible habitat). *All completions live in the admissible host category: filtered nuclear–Tate complexes with a declared BV coefficient dual (distinct from the strict dual), declared operation and kernel lists, and every inverse limit taken as a Roos complex with $\varprojlim^1$ tracked as obstruction data. The verified carrier is the weighted Tate envelope B_w , $w(d) = q^d$, $q > 1$, the minimal complete filtered subspace closing both Costello’s heat-kernel BV calculus and the Schouten bracket; its Casimir $C_{\mathfrak{h},w} = \sum_d \sum_i (w(d) e_{d,i}) \otimes (w(d)^{-1} e_d^i)$ converges, while the strict product–sum pair is a proved non-object: the diagonal coevaluation diverges.*

Three operads are kept distinct -- topological little disks E_2^{top} , the holomorphic factorization operad $E_2^{\text{ch},\text{hol}}$ in complex dimension two, and the mixed Morita-centre structure E_2^{HT} -- with no identification asserted without a topologisation or restriction datum. Four deformation parameters are kept distinct: $\hbar_{\text{QME}}$ (Costello loops), $\hbar_{\text{W}}$ (Moyal–Weyl–Capelli), $\hbar_{\omega}$ (equivariant), $\hbar_{\text{Rees}}$. Four map symbols are kept distinct: the generator boundary value, the Hochschild-centre lift (a Maurer–Cartan datum, not a construction), the proved coordinate isomorphism, and the quantum map (never asserted). The native singular kernel is two-dimensional: $K_{\Delta}^{(2)} = d\zeta_1 d\zeta_2 / (\zeta_1 \zeta_2)$; a curve vertex algebra exists only after a restriction datum with a transverse boundary condition -- without the restriction functor, no vertex algebra is constructed.

3 The theorems

Theorem 3.1 (The formal–Darboux trace-sector stalk; proved). *Under the coordinate tier (exact continuous duality, perfect residue pairing, continuity of bracket and coadjoint action), the continuous local-observable algebra of the formal current stalk is $C_{\text{CE},\text{cont}}^\bullet(\mathfrak{g})$; the closed-to-open boundary value at finite N is $c_f \mapsto \theta_f$, $u_f \mapsto J(f) \cdot (-)$ with $\theta_f(\phi_1) = -\partial_{z_2} f$, $\theta_f(\phi_2) = \partial_{z_1} f$; and after scalar reduction, stable trace passage and the Roos convention this assignment is a dg-commutative isomorphism*

$$C_{\text{CE},\text{cont}}^\bullet(\mathfrak{g}) \xrightarrow{\sim} (\text{PV}_{\text{adm}}(A_{\partial,H}^{\text{st}}), d_\pi) \xrightarrow{\text{HKR}_{\text{adm}}^{\text{cont}}} \text{HH}_{\text{adm},\text{HKR}}^\bullet(A_{\partial,H}^{\text{st}}, A_{\partial,H}^{\text{st}}).$$

Under the kernel tier (the weighted Casimir and Köthe seminorm estimates) the arity-two singular product holds: $J_f(z) J_g(w) \sim K_{\Delta}^{(2)}(z, w) (J_{\{f,g\}}(w) + \hbar_{\text{W}} N[\{f, g\}]_{\circ} \mathbf{1})$. The centre tier and the quantum tier are extra data, not consequences. The target is the admissible stable-trace Hochschild model -- explicitly not identified with the chiral Hochschild complex of the curve volumes.

Theorem 3.2 (Uniqueness and rigidity of the trace coordinate; proved). *The category of continuous maps $\tilde{J}: \mathfrak{h} \rightarrow A_\partial$ satisfying stable GL_N -invariance in matched windows, the Dirac constraint (descent modulo $[\phi_1, \phi_2]$ through the explicit homotopy), scalar reduction, Darboux naturality under formal symplectomorphisms, linear normalisation $\tilde{J}(z_i) = \overline{\text{Tr } \phi_i}$, and primitive block-sum additivity, has exactly one object: $J(f) = \lim_{\leftarrow W} J_{\leq W}(f \bmod \mathfrak{m}^{W+1})$. Dropping any hypothesis admits counterexamples. The stable window contract is quantitative: the trace identification is valid exactly for $N \geq W$ (word degree), and with a length- L Loday–Quillen–Tsygan primitive projection for $N \geq \max(W, L + 2)$; the first finite- N residue appears at $N = 2$, degree 3, by Cayley–Hamilton.*

Theorem 3.3 (The brane algebra and the one- ψ homology; proved). *The stable brane operator algebra is $\mathbf{S}(\bar{A} \oplus \bar{A}_{\text{desc}}[1])$ -- Berest–Khachatryan–Ramadoss with $\overline{HC}_o(\mathbb{C}[x, y]) = \bar{A}$, $\overline{HC}_1 \cong \bar{A}_{\text{desc}}$, higher cyclic homology vanishing. The new proof computes the one- ψ Koszul homology directly: H_1 is one-dimensional on the cyclic word class $\Psi_{k,l} = \sum_w \text{Tr}(\psi w)$, established by a cyclic-word cell complex and an equivariant Abel map to the circle with convex-polytope fibres; the higher- ψ homology vanishes in degrees ≥ 2 .*

Theorem 3.4 (The Capelli quartet; proved). *The scalar class has four proved faces. Closed: $[\bar{c}]$ is the extension class of $o \rightarrow \mathbb{C} \cdot 1 \rightarrow R \rightarrow \mathfrak{h} \rightarrow o$. Open: $\{\text{Tr } \phi_1, \text{Tr } \phi_2\} = N$, the $U(1)$ centre with charge N . Moyal: the star-commutator on trace generators truncates, $[X, Y]_\star = -\hbar_W N$ modulo the moment ideal -- the projective identity, not the impossible matrix equation $[\phi_1, \phi_2] = \hbar I$. Anomaly: $\kappa_{\text{Cap}}(J) = \hbar_W N[\bar{c}]$ in continuous Lie cohomology, with four representative constructions proved to agree. The interpretation of this class as determinant-line or modular-line curvature requires a line, a connection, and an Atiyah-class comparison that are not constructed; the Quillen curvature of the determinant line on the trace chart is a named open problem. The $\mathfrak{gl}(N|N)$ supertrace kills the class -- the Costello–Li cancellation, re-proved locally.*

Theorem 3.5 (The obstruction calculus; proved case division). *For a transported local datum in a filtered L_∞ target there are four mechanisms -- strict coherence, counterterm coherence, projective coherence (central inexact class, solved in a central extension: the Capelli case), and target replacement -- a case division, promoted to a partition only after the comparison-unifying datum is supplied. Six native obstruction classes are typed, two of them proved nonzero: the Capelli class, and the pro-Matlis window leak $z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{o,v}$, which survives every finite Matlis cutoff and forces the pro-object replacement tower. A third structural obstruction: $\mathfrak{h}$ contains $\mathfrak{sl}_2$ with $[\mathfrak{sl}_2, \mathfrak{sl}_2] = \mathfrak{sl}_2$, so lower-central pronilpotence fails on all of $\mathfrak{h}$ -- the bar–cobar route requires the cubic nilpotent truncation or a relative model.*

Theorem 3.6 (The Tate tower; proved on its stated windows). *The weighted envelope B_w is the minimal admissible completion closing the BV calculus (with heat-kernel propagator $P_{\epsilon,L}^w$ and $[Q, P_{\epsilon,L}^w] = K_\epsilon^w - K_L^w$); the cubic-ideal nilpotent truncation is honestly Tate-pronilpotent and carries the truncated Chevalley–Eilenberg–polyvector isomorphism; the bar–cobar adjunction between conilpotent Tate coalgebras and augmented Tate algebras is Quillen, a derived equivalence on the admissible subcategory; Costello’s Hadamard parametrix extends graph-by-graph to the Mittag–Leffler limit with uniform wave-front bounds (summability over graph topologies excluded, a separate obstruction); and every object of the protected-Hamiltonian moduli admits a formal-Darboux normal form, unique up to the Darboux torsor. A Costello–Li-type BV-vanishing theorem for this endpoint requires five named data, of which the brane-defect scalar contact term is decided by one of four cancellations, including the $\mathbb{C}^{N|N}$ supertrace.*

Theorem 3.7 (The global criterion; near-definitional, stated as such). *The open–closed comparison OCA_N exists at morphism stage if and only if the ten-row typed equivalence datum exists -- Swiss-cheese Maurer–Cartan lift, Morita centre lift, admissible Hochschild-centre lift, flat Darboux transport, cotangent-current realisation, Weiss–Ran descent, the order-indexed quantum master-equation tower, anomaly and locality null-homotopies, secondary primitives, and a contracting homotopy of the comparison cone. The trace-sector stalk theorem contributes exactly one row. The value of the criterion is its organisation -- the proof-order directed graph and the separation of morphism stage from cone stage -- not an existence claim.*

Conjecture 3.8 (The three compact conjectures). (i) *The chiral E_d Deligne conjecture at $d \geq 2$: the derived chiral E_d -centre carries a chiral E_{d+1} -structure -- the single bottleneck through which every global statement funnels; $d = 1$ is the Beilinson–Drinfeld case, proved in Volume I.* (ii) *The Hall–Drinfeld–Borcherds spectral sequence on compact Calabi–Yau threefolds, whose graded at $K_3 \times E$ contains the five-tuple $(0, 0, 3, 5, 24)$.* (iii) *The operator modular lift: an operator-valued cocycle on $K_3 \times E$ whose protected trace is Φ_{10} , the genus-two weight-ten section of the Hodge determinant line $\Omega_{\text{central}} = \det \pi_* \Omega^1$ on $\overline{\mathcal{A}}_2$, with $Z_{\text{BPS}}^{K_3 \times E} = \Phi_{10}^{-1}$ and $\Delta_5 = \Phi_{10}^{1/2}$ on the paramodular branch.*

The frontier volume attached to the manuscript keeps the enumerative world at arm’s length and correctly attributed: the quintic Donaldson–Thomas/stable-pairs factorisation $Z_{\text{DT}} = M(-q)^\chi Z_{\text{PT}}$ with $\chi(Q) = -200$ and M the MacMahon function is an external theorem; the Gopakumar–Vafa expansion with integral n_β^g is a constraint; and the volume’s own proposal is structural: the Maulik–Nekrasov–Okounkov–Pandharipande correspondence as a centre-lift automorphism problem -- a point of the homotopy fibre of $\text{Aut}_{E_2}(Z_{E_2}(\mathcal{A}_Q))$ over the automorphisms of the three trace normalisations, at $-q = e^{iu}$.

4 The dictionary

The local dictionary in one table:

Bulk (closed)	Brane (open)
Hamiltonian $f \in \mathfrak{h}$	gauge parameter; θ_f Hamiltonian vector field
ghost c_f (degree 1)	θ_f (degree 1)
current u_f (degree 0)	trace $J(f) = \text{Tr } f(\phi_1, \phi_2)$
cotangent tower $\rho_{a,b}$	principal parts at the brane
bracket $\{f, g\}$	$\{J(f), J(g)\} = J(\{f, g\}) + N[\{f, g\}]_0$
removed scalar line	Capelli anomaly $\hbar_W N[\bar{c}]$
singular kernel $K_\Delta^{(2)}$	arity-two operator product

The load-bearing formulas. The arithmetic kernel on monomials, $\{O_{k,l}, O_{m,n}\} = (kn - lm) O_{k+m-1, l+n-1}$ -- the w_∞ -shadow of the trace sector, the same wedge algebra whose celestial avatar appears in Volume II. The Moyal expansion on the brane truncates to odd orders with coefficients $\hbar_W^{2s+1} / (2^{2s} (2s+1)!)$. The augmentation of the one- ψ class is $\binom{k+l}{k}$. The window contract $N \geq \max(W, L+2)$. The Matlis leak $-(N+2)\rho_{0,1}$. And the terminal conjecture’s normalisations, matched to the Igusa companion: $\Phi_{10}^{\text{un}} = \Delta_5^2$, $Z_{\text{OP}} = -4096 \Delta_5^{-2}$, $\kappa_{\text{BKM}}(\Delta_5) = c_1(0)/2 = 5$.

5 The garden

One logarithmic 1-form generates the programme. On a smooth complex curve X , the form $\eta = d \log(z_1 - z_2)$ is the unique coordinate-free 1-form with a residue along the diagonal of $\text{Conf}_2(X)$; the Arnold relation generates all relations among its higher companions. From η one builds, for an augmented chiral algebra $\mathcal{A}$ on X , the ordered bar complex $\overline{B}_X^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}}) \otimes \Omega_{\log}^\bullet(\overline{\text{FM}}_\bullet(X))$; its differential squares to zero exactly when $\mathcal{A}$ satisfies the Borchers identity. Assembled over $\overline{\mathcal{M}}_{g,n}$, the total differential is a Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_o$; genus ≥ 1 flatness fails by the curvature $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \omega_g$, with $\kappa(\text{Vir}_c) = c/2$, $\kappa(V_k(\mathfrak{g})) = \dim \mathfrak{g} (k + h^\vee)/2h^\vee$ the central charge read as modular curvature; the Verdier–Koszul dual obeys the conductor laws $c(\mathcal{A}) + c(\mathcal{A}^\dagger) = K$ with $K(\text{Vir}) = 26$, $K(\mathcal{W}_N) = 4N^3 - 2N - 2$, $K(V_k(\mathfrak{g})) = 2 \dim \mathfrak{g}$.

Every object sits on one chain: primitive factorization category $\rightarrow$ chart algebra $\rightarrow$ bar shadow $\rightarrow$ derived chiral centre (the bulk) $\rightarrow$ acting line operators $\rightarrow$ scalar modular shadow. Identifying stages without the licensing datum -- shadow = object -- is the master error; this volume is the chain’s local microscope, and its seventeen-site catalogue of exactly such collapses, each with its corrected form, is the sharpest statement of the discipline anywhere in the programme.

The five gardens are five readings of the chain. *Volume I* (chiral-bar-cobar) builds the engine on algebraic curves -- in particular the $d = 1$ chiral Deligne input that this volume’s central conjecture generalises. *Volume II* (chiral-bar-cobar-vol2) reads the bar complex as BV/BRST physics on $\mathbb{C} \times \mathbb{R}$: its curve–line theorems apply here only after the named restriction datum, since the native geometry is $\mathbb{R}^2 \times \mathbb{C}^2$. *Volume III* (calabi-yau-quantum-groups) is the compact target world -- its two-stage functor and Hall doubles are exactly what the typed firewalls of this volume keep at criterion distance until the compact data exist. *This volume* computes the local germ: the trace coordinate at N Dirac branes, its uniqueness, its anomaly, and the exact list of data separating germ from globe -- including the holomorphic de Rham obstruction $H^1(K_3 \times E, \mathcal{O}) \neq 0$ that blocks the naive local-to-global route. *The Igusa cusp form* (igusa-cusp-form) is the terminal scalar this volume’s third conjecture points at: Δ_5 as the genus-two section of the modular line whose operator lift is the shared frontier.

The single question underneath: characterise the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ as a geometric object. The proposed answer -- endomorphisms of a Lagrangian in a (-2) -shifted symplectic stack of vacua -- has Volume I computing its Taylor jets, Volume II its local composition law, Volume III its geometric source, and this volume its formal-local microscope: the one stalk where every structure is computed to the end, so that the global obstructions can be named rather than guessed.

6 The frontier

In order of depth:

- (i) The chiral E_d Deligne conjecture at $d \geq 2$ -- the single bottleneck; all ten rows of the global datum presuppose it.
- (ii) The all-order quantum master-equation tower for the brane-defect coupling, with the anomaly and locality null-homotopies -- the quantum tier the stalk theorem deliberately excludes.
- (iii) The determinant-line problem: construct the line, connection and Atiyah comparison that read the Capelli class $\hbar_{\mathbb{W}} N[\tilde{c}]$ as curvature; then the Quillen curvature of the trace chart.
- (iv) The operator lift of Φ_{10} -- shared with the Igusa volume and Volume II’s gravity-line construction problem; the modular line Ω_{central} is defined, its operator-valued cocycle is not.

- (v) The Maulik–Nekrasov–Okounkov–Pandharipande centre-lift: the correspondence as a canonical E_2 -centre automorphism, with cohomological-Hall orientation data as prerequisite.

What stands unconditionally: the stalk isomorphism with its uniqueness axioms; the brane algebra and one- ψ homology; the Capelli quartet with its four faces and its supertrace cancellation; the weighted-Tate completion theory with its proved non-objects and window leaks; and the typed ledger itself -- the volume's real invention, which converts "local implies global" from a slogan into a finite list of named data.